

Measuring subnational democracy with Varieties of Democracy (V-Dem) survey data

An application using the case of Colombia

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- Then, we **weight** these factors according to the levels of V-Dem inter-expert agreement.

*The results can be interpreted as **expected levels** of subnational democracy based on theoretical knowledge and observational data.*

Introduction

Observational vs. Expert data

Observational data is often used in democracy studies, but below the national-level this information is often scarce, non-standardized, expensive to obtain, and only partially informative.

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Expert data may compensate for these limitations because:

- information is absorbed from multiple sources (e.g., reading, following news, conversations, personal experiences)
- experts are less likely to be deceived by insincere statements or to consider one attribute definitive when other conditions render it irrelevant

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Existing comparative measures rely only on **observational** data rather than expert data (Pérez Sandoval 2023), while **individual** country measures are not easy to reproduce across contexts (Hernández Valdez 2000; Ortega and Somuano in Meixueiro and Pérez 2014; Gervasoni 2018; and Grumbach 2023).

Our measurement approach

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- areas where elections are more free and fair
- areas where civil liberties are stronger
- areas where civil liberties are weaker

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Each question includes **twenty-two (22)** “choose all that apply” variables, providing a range of generic concepts applicable to any country context.

V-Dem's subnational data (cont.)

<i>How would you describe the areas of the country in which elections are significantly less free and fair? [v2elsnlfc]</i>
<i>How would you describe the areas of the country in which elections are significantly more free and fair? [v2elsnmrfc]</i>
<i>How would you describe the areas of the country where government officials' respect for civil liberties is significantly stronger? [v2clrgstch]</i>
<i>How would you describe the areas of the country where government officials' respect for civil liberties is significantly weaker? [v2clrgwkch]</i>
Clarification: Choose all that apply.
0: Rural. (0=No, 1=Yes)
1: Urban. (0=No, 1=Yes)
2: Areas that are less economically developed. (0=No, 1=Yes)
3: Areas that are more economically developed. (0=No, 1=Yes)
4: Inside the capital city. (0=No, 1=Yes)
5: Outside the capital city. (0=No, 1=Yes)
6: North. (0=No, 1=Yes)
7: South. (0=No, 1=Yes)
8: West. (0=No, 1=Yes)
9: East. (0=No, 1=Yes)
10: Areas of civil unrest (including areas where insurgent groups are active). (0=No, 1=Yes)
11: Areas where illicit activity is widespread. (0=No, 1=Yes)
12: Areas that are very sparsely populated. (0=No, 1=Yes)
13: Areas that are remote (difficult to reach by available transportation, for example). (0=No, 1=Yes)
14: Areas where there are indigenous populations. (0=No, 1=Yes)
15: Areas where the national ruling party or group is strong. (0=No, 1=Yes)
16: Areas where the national ruling party or group is weak. (0=No, 1=Yes)
17: Areas that were subject to a longer period of foreign rule. (0=No, 1=Yes)
18: Areas that were subject to a shorter period of foreign rule. (0=No, 1=Yes)
19: Areas that were recently subject to foreign rule. (0=No, 1=Yes)
20: Areas that have not recently been subject to foreign rule. (0=No, 1=Yes)
21: None of the above. (0=No, 1=Yes)

Figure 1: V-Dem's 4 subnational questions and the 22 "choose all that apply" responses

Rendering V-Dem data useful for subnational analysis

While many variables are available (4*22 inter-expert ratings), these are binary options without specific indicators.

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1. **Conceptualization:** selection of within-country indicators for each V-Dem variable
2. **Operationalization:** data reduction and weighting using V-Dem responses
3. **Measurement:** estimation and visualization of weighted average subnational democracy scores

Measurement approach (flow-chart)

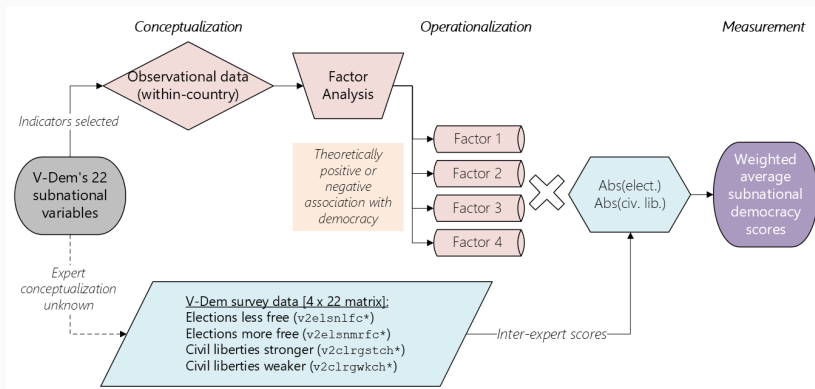


Figure 2: Our approach combines V-Dem expert data with observational data using data reduction techniques and weighting to measure subnational democracy scores.

Conceptualization

Selecting indicators

We do not know how experts conceptualized the V-Dem variables in their responses, so we identify specific indicators (mostly **continuous**) from official state data or others' scholarly research to measure V-Dem variables.

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ID	Variables	Indicator
0	Rural	Rurality Index (DANE)
1	Urban	
2	Less econ. devt.	Mun. Value Added / Multi-dimensional Poverty Index (DNP)
3	More econ. devt.	
4	Inside capital	Geographic distance from Bogotá (DANE)
5	Outside capital	
6	North	Macro-regions (DANE)
7	South	
8	West	
9	East	
10	Civil unrest	Forced displacement / Confinements (Min. Defense)
11	Illicit activity	Robberies / Homicides / Manual eradication (Min. Defense/UNODC)
12	Sparse population	Population density (DANE)
13	Remote	Distance to market (CEDE)
14	Indigenous pop.	Indigenous / Ethnic populations (DANE)
15	Ruling party strong	Presidential margin of victory (RNEC)
16	Ruling party weak	

Figure 3: Indicators selected for Colombia 2018

Operationalization

Data reduction: Factor analysis

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As a result, we analyze 13 of the 22 variables, using 17 indicators in total.

Factor analysis: Colombia 2018

Factor Analysis Loadings

	MR1	MR2	MR4	MR3
Rurality (0-1)	-0.127	-0.592	0.325	0.023
VAM (2-3)	0.268	0.844	-0.340	0.174
IPM (2-3)	0.217	-0.409	0.834	0.328
Distance Bog. (4-5)	0.474	0.154	0.696	0.162
Displaced (10)	0.289	0.235	0.103	0.603
Confinement (10)	0.076	-0.046	0.124	0.119
Robberies (11)	0.045	0.572	-0.589	0.016
Homicides (11)	0.239	0.237	0.013	0.625
Eradication (11)	0.192	0.015	0.229	0.601
Pop. Density (12)	-0.020	0.596	-0.408	-0.342
Total Pop. (12)	0.450	0.940	-0.056	0.230
Distance Market (13)	0.345	-0.190	0.434	0.125
Indigenous (14)	0.959	0.457	0.245	0.312
Ethnic (14)	0.881	0.661	0.337	0.314
Indig. per cap. (14)	0.937	0.190	0.313	0.295
Ethnic per cap. (14)	0.884	0.307	0.465	0.301
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- **MR3:** Areas with victims of violence and civil unrest (#10, 11)

Data reduction: Colombia 2018 (cont.)

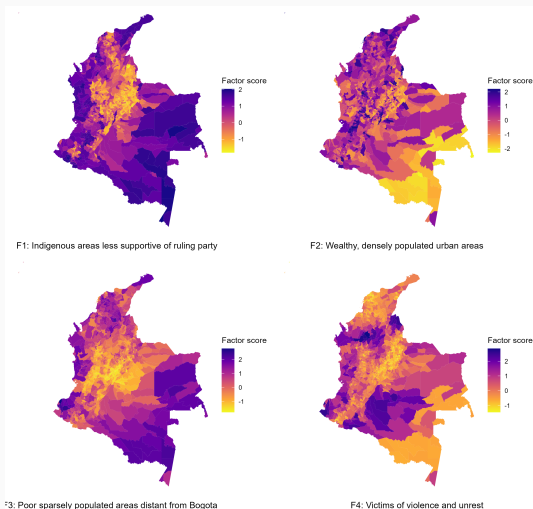


Figure 4: Factor scores mapped at the municipal-level

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The factors suggest **splits** along rural and urban, poorer and richer, and perhaps areas dominated by indigenous and non-indigenous populations.

*In terms of democratic theory, these splits roughly align with **modernization** arguments that urban, wealthy, and ethnically diverse areas are associated with higher levels of democracy.*

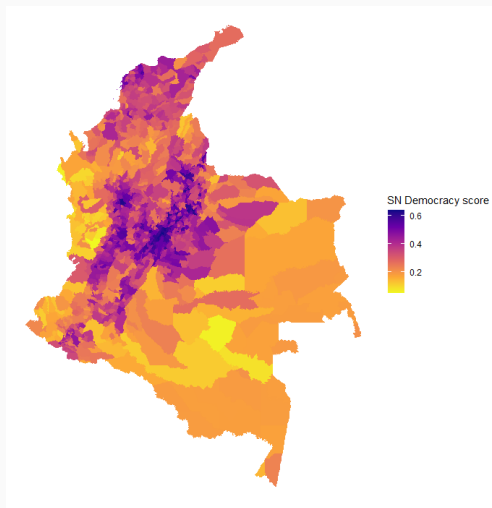
Weighting FA by V-Dem scores

We weight the factors by taking the **absolute value** of the proportions across both “less/weaker” and “more/stronger” categories in the V-Dem data.

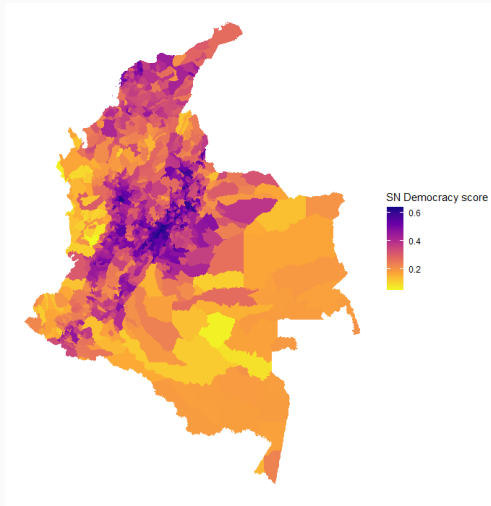
Variable	EL Less free	EL More free	CL Stronger	CL Weaker	Abs(EL)	Abs(CL)
Rural	0.875	0.000	0.000	1.000	0.875	1.000
Urban	0.125	1.000	0.889	0.000	0.875	0.889
Less econ devt	0.875	0.111	0.000	1.000	0.764	1.000
More econ devt	0.125	0.667	1.000	0.000	0.542	1.000
Inside capital	0.000	0.889	1.000	0.000	0.889	1.000
Outside capital	0.375	0.000	0.000	0.600	0.375	0.600
North	0.250	0.222	0.333	0.100	0.028	0.233
South	0.250	0.000	0.000	0.500	0.250	0.500
West	0.125	0.000	0.111	0.300	0.125	0.189
East	0.125	0.000	0.111	0.300	0.125	0.189
Civil unrest	1.000	0.000	0.000	0.900	1.000	0.900
Illicit activity	0.875	0.000	0.000	0.900	0.875	0.900
Sparse pop.	0.375	0.000	0.000	0.600	0.375	0.600
Remote	0.875	0.000	0.000	0.800	0.875	0.800
Indigenous	0.500	0.000	0.000	0.900	0.500	0.900
Ruling party strong	0.000	0.222	0.333	0.000	0.222	0.333
Ruling party weak	0.000	0.000	0.000	0.100	0.000	0.100

Measurement

Municipal-level scores: Colombia 2018

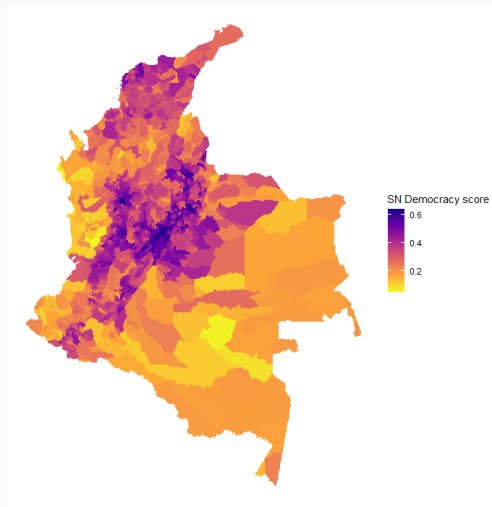


Municipal-level scores: Colombia 2018



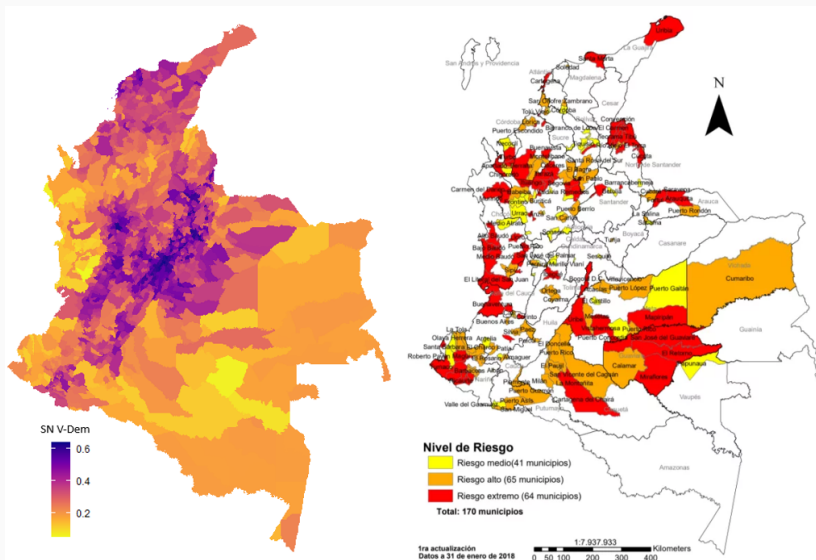
- Weighted averages show higher democratic clusters (blue) in urban areas along the Andean mountain ranges and less so along the Caribbean coast.

Municipal-level scores: Colombia 2018



- Weighted averages show higher democratic clusters (blue) in urban areas along the Andean mountain ranges and less so along the Caribbean coast.
- Weaker democratic clusters (yellow) are visible in rural areas and somewhat on the Pacific coast.

Face validity? Comparison with 2018 electoral violence (MOE)



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Our analysis of Colombia is a **proof-of-concept**, leveraging well-developed geographic data from DANE and other sources.

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Subnational democracy can be measured using V-Dem data related to electoral and civil liberties dimensions of democracy.

- Results can be interpreted as **expectations** for subnational democracy based on theoretical knowledge and observational data.
- Some expert-level ratings may be inaccurate, but if theoretical assumptions are true, this provides a first step in using V-Dem data for subnational analysis.

Our analysis of Colombia is a **proof-of-concept**, leveraging well-developed geographic data from DANE and other sources.

- Provisional results are based on limited subjective information, but also conditioned by levels of inter-coder agreement.

Next steps

It is necessary to **validate** our preliminary results using:

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Also, to improve the **internal validity** of the method, we can:

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- anchor ratings to V-Dem's national scores to establish **ranges of variation** (i.e., Clean Elections Index could anchor subnational elections scores)

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- use **Bayesian analysis** of priors around V-Dem's 22 subnational variables and posterior expert ratings
- extend to other countries with subnational measures (e.g., Argentina, Mexico, India, USA, Russia)

Thank you!

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